TECHNICAL REPORT

MULTI-OBJECT DETECTION AND TRACKING USING YOLOv8 AND DEEPSORT

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1. Introduction

This project implements a computer vision pipeline for detecting and tracking multiple objects in a video stream. The system uses YOLOv8 for object detection and DeepSORT for tracking. Each detected object is assigned a unique ID, which is maintained consistently across frames. This enables real-time tracking of multiple objects with identity preservation.

2. Model Used

YOLOv8 (You Only Look Once version 8) is used for object detection. It is a state-of-the-art deep learning model known for its high accuracy and real-time performance. The model detects objects in each frame and provides bounding boxes along with confidence scores.

3. Tracking Algorithm

DeepSORT (Deep Simple Online and Realtime Tracking) is used for tracking detected objects. It assigns a unique ID to each object and maintains that identity across frames.

DeepSORT works using:

- Kalman Filter for motion prediction

- Appearance feature extraction for matching objects

- Hungarian Algorithm for data association

4. System Justification

YOLOv8 provides fast and accurate object detection, while DeepSORT ensures identity consistency across frames. This combination is widely used in real-world applications such as surveillance systems, traffic monitoring, and people tracking.

5. System Pipeline

The system follows the steps below:

1. Input video is captured frame by frame
2. YOLOv8 detects objects in each frame
3. Bounding boxes and confidence scores are extracted
4. Detections are passed to DeepSORT
5. DeepSORT assigns unique IDs to objects
6. Bounding boxes and IDs are drawn on frames
7. Final annotated output video is generated
8. ID Consistency

DeepSORT maintains identity consistency using:

- Motion prediction through Kalman Filter
- Appearance-based feature matching
- Data association using Hungarian Algorithm

This ensures that the same object retains the same ID even when it moves across frames.

7. Challenges Faced

- Occlusion (objects overlapping each other)
- Fast movement of objects
- Motion blur in video
- Similar-looking objects

8. Failure Cases

- ID switching when objects overlap heavily
- Missed detections in low-quality frames
- Reduced accuracy in crowded scenes

9. Future Improvements

- Use a custom-trained YOLO model for better accuracy
- Improve tracking using advanced re-identification models
- Use higher resolution videos for better detection
- Optimize performance for real-time deployment

10. Conclusion

This project successfully demonstrates real-time multi-object detection and tracking using YOLOv8 and DeepSORT. The system is capable of maintaining identity consistency across frames and performs effectively in standard conditions. With further improvements, it can be extended for use in real-world applications such as surveillance, traffic analysis, and smart monitoring systems.